

ORIGINAL ARTICLES

A Model for the Computer Analysis of Synchronous DNA Distributions Obtained by Flow Cytometry¹

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A new model for obtaining the distributions of cells in the cell cycle by computer analysis of flow cytometry data is proposed. The model is able to analyze distributions from both asynchronous and synchronous cell populations. A culture of HeLa cells is pulse-labeled with tritiated thymidine at various times after synchronization by the mitotic selection method, then analyzed by

flow cytometry and autoradiography. The results show good agreement between the percentage S-phase cells determined by the model and by autoradiography.

Key terms: Computer model, synchronous, flow cytometry, autoradiography, mitotic selection

Flow cytometry has become an important tool for research in cell biology (4, 15, 18, 21, 23). An important application of flow cytometry is to determine the distribution of DNA content per cell in a population of cells at a very rapid rate and with good statistics. However, the important parameters usually desired—the proportion of cells in each phase of the cell cycle—cannot be accurately determined without using a computer to model the distribution. Various mathematical models have been proposed to accomplish this, with varying degrees of success. These have been reviewed recently (12).

These models may be categorized by two basic conceptual approaches. One approach is to develop a function that will accurately represent the G_1 , S and $G_2 + M$ components of the DNA histogram derived from flow cytometry, and thus obtain the relative proportion of cells in each phase (1, 2, 5, 7, 8, 9). The other approach is more basic in the sense that it models the cell age distribution and then transforms that distribution into a DNA distribution (10, 11, 17, 24), usually by matrix manipulation. This approach requires knowledge of the rate of DNA synthesis during S-phase. It does allow the capability of determining additional information, such as the phase times, and in fact the percentage of cells in each phase is not usually the primary information desired from these models.

The model presented here is based on the first approach.

Dean and Jett (5) presented the first serious attempt to model accurately the DNA distributions of asynchronous cell populations. With such populations, their model gives excellent results. The model fits normal (Gaussian) distributions to the G_1 and $G_2 + M$ peaks, and a broadened second-order polynomial to S-phase. The broadening accounts for dispersion from a variety of causes, including instrumental, staining and biologic factors. The limitation of the model is that it can only fit distributions where a second order polynomial can accurately represent the S-phase data. This leaves out nearly all synchronous populations.

Fried (7-9) presented another model in which a series of normal curves is fitted to S-phase. The advantage of this model is that it can fit synchronous populations, with varying degrees of success. However, the model is very sensitive to the resolution of the data. Furthermore, it requires considerable operator monitoring to obtain proper parameters (12). The main difficulty is in choosing the spacing of the S-phase compartments. If this is not chosen carefully, the resulting fit can have large fluctuations in the estimated number of cells from one compartment to the next. The spacing is correlated with the coefficient of variation of the G_1 peak (8). This is a rather arbitrary factor because ideally there should be one compartment per channel since DNA synthesis is a continuous rather than discrete process.

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The model proposed here is similar to the Dean and Jett approach, in that a smooth curve is fitted to the S-phase region. The important advantage of this model is that it can fit data from synchronous, as well as asynchronous populations. This is achieved by using a single normal curve riding on a second order polynomial in the S-phase region. This gives the function enough flexibility to fit data from synchronous populations. Each channel in S-phase is then broadened by a normal distribution.

Synchronizing cells by the mitotic selection procedure (22) is a common practice in cell biology. An important application of flow cytometry is to analyze the distribution of cells at various times after synchronization in mitosis and to study the effects of various agents on these distributions. A stringent test of a model is its ability to analyze these distributions, since they encompass virtually all combinations of cells in the various life cycle phases.

To test the ability of this new model to analyze synchronous populations, a HeLa S3 culture was synchronized and samples were prepared for labeling index and flow cytometry at various times throughout a complete cell cycle. These data were analyzed with this model, and the labeling index was determined by (tritiated thymidine ^3H -TdR) uptake. The total fit by the model to the data was good in every case, and the S-phase fractions determined from the model and from autoradiography were in good agreement.

Description of Model

Algorithm: The general form of the model is very similar to that given by Dean and Jett (5) and Lindmo and Aarnaes (19). The algorithm is composed of three parts: a normal distribution for the G_1 peak (F_1); a normal distribution for the $G_2 + M$ peak (F_2); and a function, $f(x_j)$, broadened by convolution with a normal distribution for the S-phase distribution (F_s). The total function is $F(x) = F_1(x) + F_s(x) + F_2(x)$, with the various functional parts defined below.

$$F_1(x) = \frac{N_1}{\sqrt{2\pi} \sigma_1} \exp \left[-\frac{(x - x_1)^2}{2\sigma_1^2} \right] \quad (1)$$

$$F_s(x) = \sum_{j=x_1}^{x_2} f(x_j) \frac{1}{\sqrt{2\pi} \sigma_1 \frac{x_j}{x_1}} \exp \left[-\frac{(x - x_j)^2}{2\left(\sigma_1 \frac{x_j}{x_1}\right)^2} \right] \quad (2)$$

$$F_2(x) = \frac{N_2}{\sqrt{2\pi} \sigma_2} \exp \left[-\frac{(x - x_2)^2}{2\sigma_2^2} \right] \quad (3)$$

Each channel in S-phase, x_j , is assumed to have the same coefficient of variation (CV) as the G_1 peak, defined as σ_1/x_1 . This CV represents broadening due to instrumental and biological factors. The $G_2 + M$ peak location (x_2) and CV (σ_2/x_2) are usually independent parameters which are obtained by the fitting algorithm. However, in cases where a large component of late-S cells dominates the $G_2 + M$ peak, these two parameters must be fixed in relation to the G_1 peak. The CV is then forced to be the same as that found for the G_1 peak, and the peak location is a constant times x_1 . The constant is usually around 1.9 for our system, instead of 2.0. The proper value to use will depend on the flow cytometer system being used and must be determined by independent methods.

The function used to represent the S-phase distribution

depends on the type of population to be analyzed. For asynchronous populations the function used is a simple second-order polynomial:

$$f(x_j) = A + Bx_j + Cx_j^2 \quad (4)$$

which is identical to the Dean and Jett model.

For populations with a more complex S-phase distribution, such as most synchronous populations, the function used is:

$$f(x_j) = A + Bx_j + Cx_j^2 + \frac{N_s}{\sqrt{2\pi} \sigma_s} \exp \left[-\frac{(x_j - x_s)^2}{2\sigma_s^2} \right] \quad (5)$$

which is a normal curve riding on a second order polynomial. The parameters of the normal curve are all fitted by the program, so its CV has no relationship to the CV of the G_1 peak. The normal curve plus the polynomial describes the distribution of cells in the S-phase compartments, which is different from the broadening of each compartment due to instrumental and biologic dispersion. The latter dispersion is accounted for by the broadening factor in equation (2), which has the same CV as the G_1 peak. The additional normal curve in equation (5) gives the model enough flexibility to be able to fit synchronous populations with complex S-phase distributions, while maintaining the advantages of using a smooth curve.

The three extra parameters in the S-phase function for synchronous populations (equation (5)) significantly increase the computing time compared to the Dean and Jett model using the second order polynomial alone (equation (4)). For this reason, equation (4) is used for all asynchronous populations. At the beginning of the analysis, the operator simply specifies which type of population is going to be analyzed and the program chooses the proper function automatically.

The parameters to be determined are: N_1 , σ_1 , x_1 (G_1 peak); N_2 , σ_2 , x_2 ($G_2 + M$ peak); A , B , C , N_s , σ_s , x_s (S-phase) when using equation 5. All 12 of these parameters are fit automatically by the program, except in the case already discussed where the $G_2 + M$ peak must be fixed in relation to the G_1 peak. In this case, σ_2 and x_2 are not freely floating but are tied to σ_1 and x_1 .

Initial input: Asynchronous populations require only the data for analysis. The program finds G_1 and $G_2 + M$ peak locations and makes initial estimates of the parameters from these peak values.

Synchronous populations may not have a prominent G_1 or $G_2 + M$ peak. In this case, the operator must input three parameters—the G_1 , $G_2 + M$ and S peak locations. There may not be a well-defined peak in S-phase, but the estimated peak location should correspond to the region of S-phase with the highest cell density. These are only initial estimates to allow the program to begin. All of the final parameters will be found by the program, using the fitting procedure.

Fitting procedure: The final parameters are determined by minimizing the sum of the squares of the weighted deviations of the model from the data (χ^2). Since the function $F(x)$ in the model is nonlinear, a true analytical least squares solution does not exist. Instead, the parameters are determined by searching in a multidimensional parameter space, according to a method described in detail by Bevington (3), using the algorithm of Marquardt (20). This method utilizes

a gradient search to approach the final parameters rapidly, then uses a quasianalytical approach to reach the final solution efficiently.

The parameters are all adjusted simultaneously according to the algorithm, then χ^2 is determined and compared with that of the previous iteration. When the change in χ^2 is less than 1%, the iterative process stops and the last set of parameters is the solution set.

After determining the optimal parameters, the percentage of cells in each phase is calculated. N_1 and N_2 give the number of G_1 and $G_2 + M$ cells directly, while the number of cells in S-phase is the summation of $F(x_i)$ from x_1 to x_2 . Since the S-phase distribution is known, subpopulations of S-phase can also be determined. In the current version, S-phase is divided into quarters and the percentage of cells in each quarter is determined. This gives more detailed knowledge of S-phase, in addition to the total number of cells.

At the operator's option, the actual data, as well as the total fit and the S-phase component, are plotted on a graphics terminal.

Materials and Methods

Cell line and growth conditions: S3 HeLa cells were used in these experiments to test the model. They were grown at 37°C in F12 medium (13) containing 10% fetal calf serum and subcultured three times a week. They have a doubling time of approximately 20 hr. The cells were synchronized by mitotic selection (22). The cells were held in mitosis at 0°C for a maximum of 1½ hr during the selection procedure. The mitotic index was scored for the pooled fractions and was 93%.

The pooled mitotic cells were centrifuged and resuspended in thymidine-free F12 medium, then plated into parallel cultures in T-25 Falcon flasks, which were incubated at 37°C in a humidified atmosphere of 5% CO₂ in air.

Autoradiography: The cells were given a 10-min pulse of ³H-TdR (2 µCi/ml; specific activity of 34 Ci/mmol) at various times after plating them in thymidine-free medium. The ³H-TdR uptake was quenched by adding a 1000-fold excess of cold thymidine after 10 min. The cells were harvested quantitatively by trypsinization, using 0.03% trypsin in saline D1 (14). The suspension was then centrifuged, decanted, and resuspended in Hanks' balanced salt solution. Part of the sample was fixed in 3:1 methanol/acetic acid after a hypotonic rinse, and slides were prepared for autoradiography. The rest of the sample was prepared for flow cytometry.

The slides were placed in 1 N hydrochloric acid at 60°C for 3 min, then rinsed thoroughly. They were coated with Kodak NTB photographic emulsion and stored at 4°C in the dark for 3 days. They were then developed with Kodak D19 for 5 min at 4°C. After staining through the emulsion with Giemsa, they were scored for labeled cells. An unlabeled control was prepared for background correction. Two independent observers scored the labeling index, defined as the fraction of cells labeled with at least three grains. A minimum of 500 cells were counted for each determination.

Staining for flow cytometry: After the cells had been rinsed in Hanks' balanced salt solution, they were centrifuged and resuspended in one part cold Hanks' balanced salt solution, followed by three parts cold 95% ethanol. The latter was added while vortexing the sample virgously to minimize clumping. The final ethanol concentration was 70%. After remaining in the ethanol fix for at least 30 min, the cells were centrifuged out and resuspended in 1.7×10^{-5} M Chromomycin A3 solution (16). After remaining in the dark at room temperature for at least 30 min, they were analyzed by flow cytometry, with the laser

set at 457.9 nm. A total of 40,000 cells were analyzed from each sample.

Instrumentation and data analysis: The flow cytometer in use was developed at Colorado State University from components provided by various manufacturers, and is described in detail elsewhere (6). It is a high resolution system, capable of measuring coefficients of variation on uniform polystyrene spheres of 1.7% or less. Data were stored in a Nuclear Data ND100 multichannel analyzer (Nuclear Data, Schaumburg, Ill.), then transferred to magnetic cassette tape on a TI 733 ASR (Texas Instruments, Houston, Texas) computer terminal for permanent storage. The data were analyzed on a CDC Cyber 172 computer in an interactive mode, and the resulting fits were displayed on a graphics terminal (Tektronic, Inc., Beaverton, Oregon).

Results

The results of fitting an asynchronous population of HeLa cells with the model are shown in Figure 1. Both equations (4) and (5) were used to fit the data, with nearly identical results (see Table 1). No detectable difference could be seen in the resulting fits. In these and subsequent figures, the *solid line* connects the actual data values, the *broken line* represents the total fit to the data and the *dashed line* represents the S-phase component of the fit. The results of fitting the wave of synchronized cells moving through the cell cycle are illustrated in Figures 2 and 3 and summarized in Table 1 and Figure 4.

The values of reduced χ^2 in Table 1 give an indication of the goodness of the fit. Reduced χ^2 is the sum of the weighted squares of the deviations of the model from the data, divided by the number of degrees of freedom. The weighting factor is assumed to be the reciprocal of the data value.

The values given for the percentage labeled cells are the mean values, and their associated standard deviations, ob-

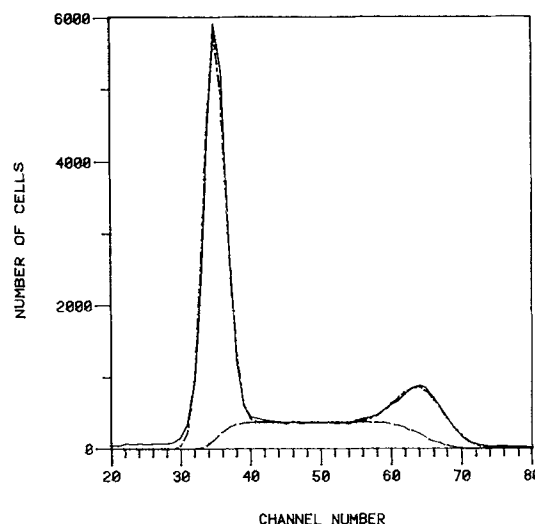


FIG. 1. Flow cytometry data obtained from an asynchronous HeLa cell population and analyzed by the model. The *solid line* connects the actual data values. (----) is the total fit by the model to the data. (----) is the S-phase portion of the total fit to the data. This representation also applies for Figures 2 and 3. A point-by-point subtraction of the S-phase curve from the total fit would yield the normal curves which represent the G_1 and $G_2 + M$ peaks.

Table 1

Figure	Time after Syn- chro- ny	Reduced χ^2	Percentage G ₁	Percentage G ₂ + M	Percentage S	Percentage Labeled Cells
	hr					
1	^a	4.1	58.1	14.2	27.7	^b
1	^a	4.3	58.1	13.7	28.2	^b
2A	7.5	9.0	86.8	9.2	4.0	4.5 ± 0.7
2B	9.0	3.7	79.3	7.3	13.4	20.0 ± 2.2
2C	11.0	4.7	46.6	6.8	46.6	55.4 ± 3.0
2D	13.0	4.8	20.2	5.5	74.2	78.6 ± 0.6
3A	15.0	2.4	12.3	6.3	81.4	86.0 ± 1.2
3B	17.0	2.5	9.2	21.6	69.2	74.4 ± 1.1
3C	18.0	4.0	9.0	34.1	57.0	63.6 ± 2.6
3D	20.0	6.1	22.0	42.7	35.4	35.6 ± 1.1

^a Cells were in asynchronous growth.
^b Labeling index was not done on this sample.

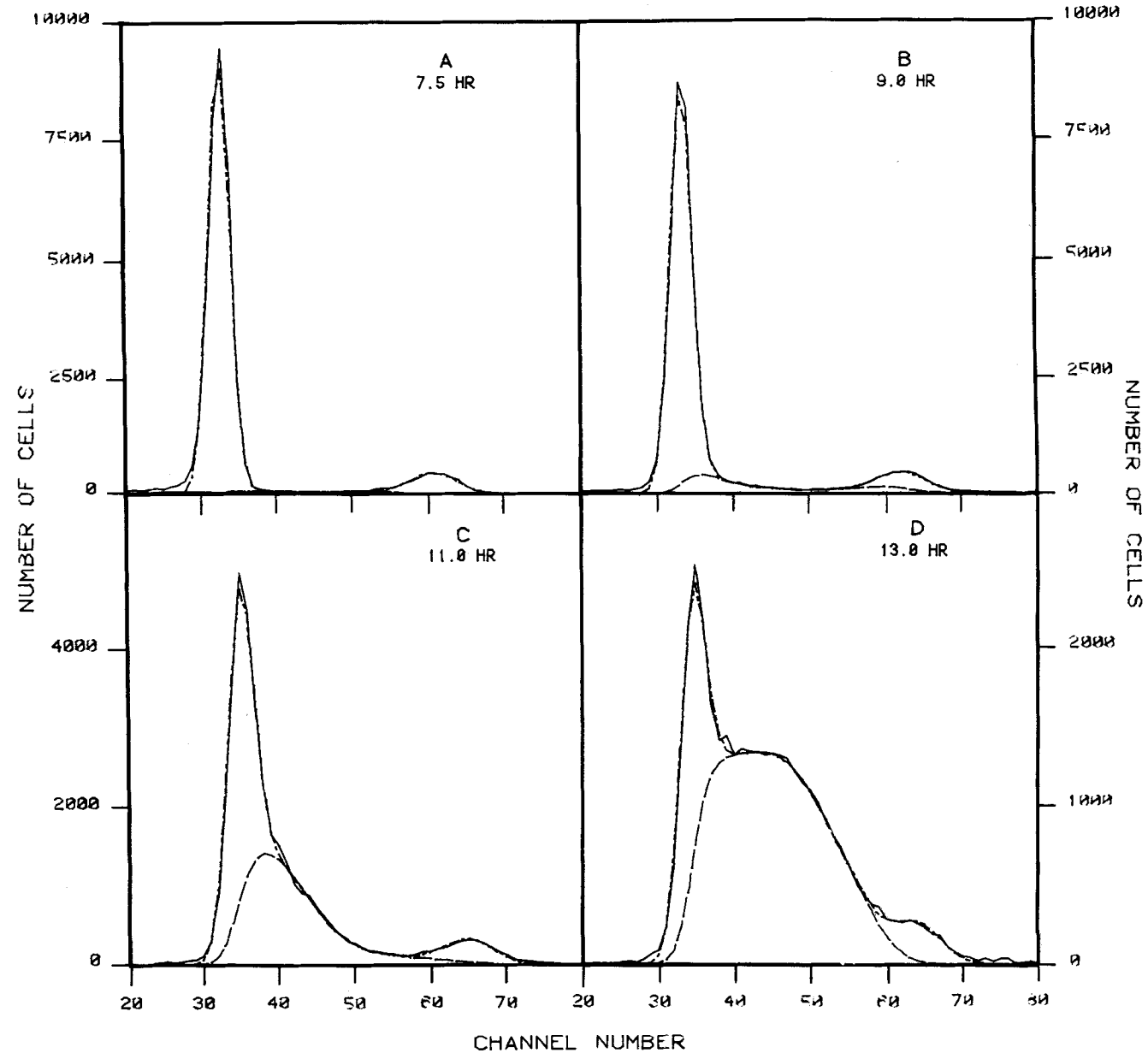


FIG. 2. Distributions of HeLa cell cultures obtained at various times after plating cells from a mitotic selection procedure. Data are analyzed by the model, using the same legend as for Figure 1. A, 7.5 hr; B, 9.0 hr; C, 11.0 hr; D, 13.0 hr.

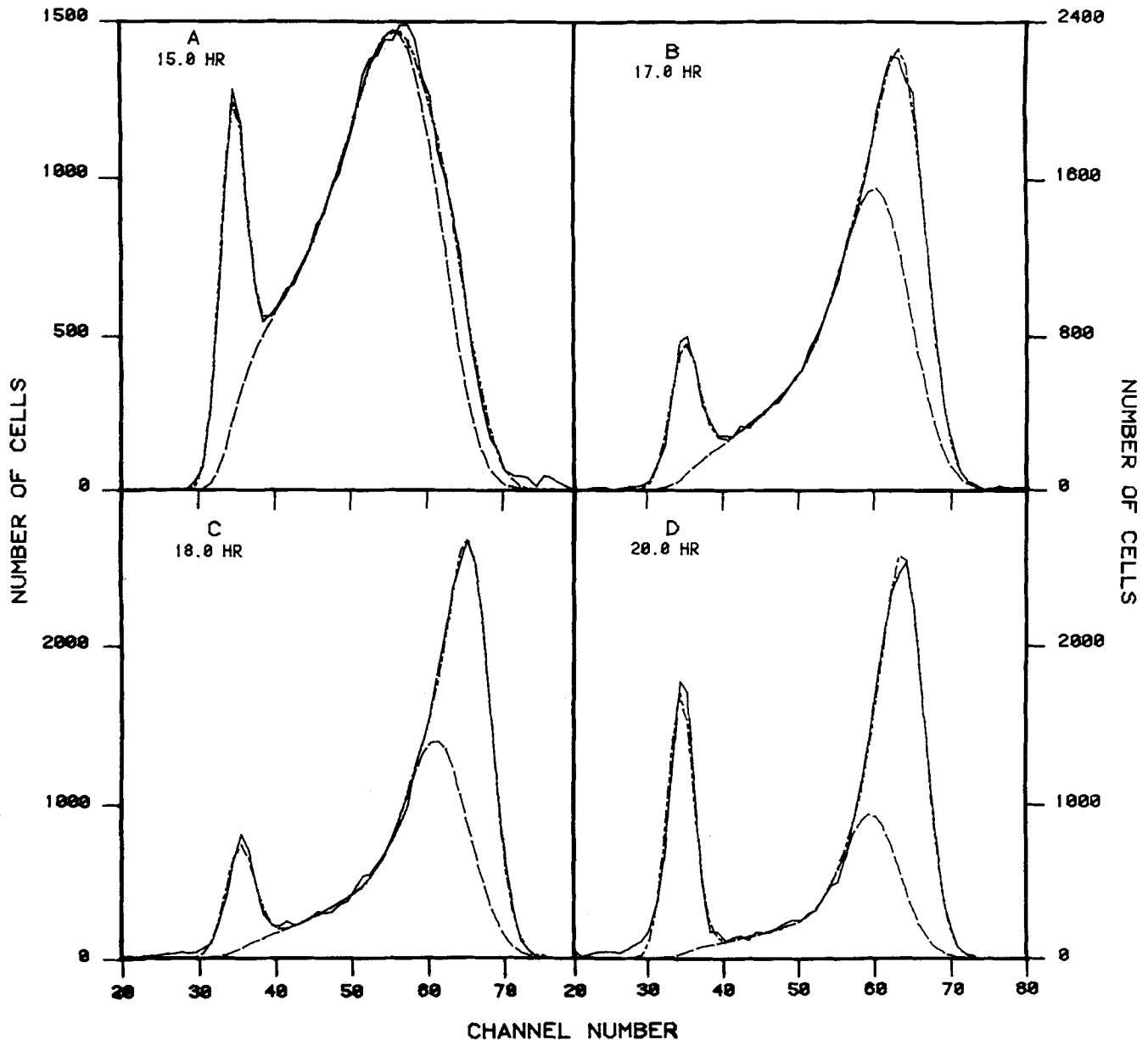


FIG. 3. Distributions of HeLa cell cultures obtained at various times after plating cells from a mitotic selection procedure. Data are analyzed by the model, using the same legend as for Figure 1. A, 15.0 hr; B, 17.0 hr; C, 18.0 hr; D, 20.0 hr.

tained by two observers scoring the labeling indices independently.

The total number of cells measured by flow cytometry was 40,000 for each distribution. When the cells are spread throughout S-phase, there will be fewer cells per channel than when the majority are in a few channels in G_1 . This is the reason the maximum number of cells in the peak channel varies for the different distributions.

Discussion

This paper has presented a new model designed to fit data obtained by flow cytometry from synchronous cell populations and to determine the percentage of cells in each phase of the cell cycle. The results of following an initially synchronous population with both flow cytometry and labeling index show

that the model fits the data well. The fit to asynchronous populations is equally good whether the Dean and Jett model or the new model is used. Although labeling index was not done for this sample, it has been done on other samples and the results compare well with the model.

The comparisons in Table 1 of the labeling index data with the S-phase determination by the model show agreement within a few percent in all cases. Figure 4 shows a direct comparison between the two determinations throughout the cell cycle. It is clear that the model is able to determine accurately the S-phase fraction of the populations for a variety of different synchronous cell cultures. Since the overall fit of the model to the data is very good, this implies that the model is also accurate in determining the G_1 and $G_2 + M$ populations.

It is interesting to note that the model consistently under-

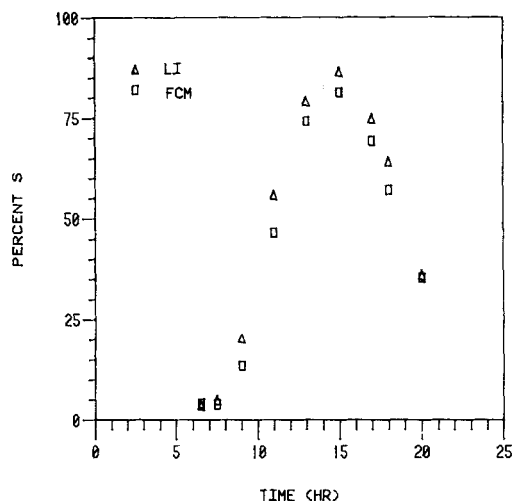


FIG. 4. Comparison of the S-phase fractions determined by the model (□) or by autoradiography (△), as a function of time after plating cells from a mitotic selection procedure.

estimates S-phase compared to the labeling index by a slight amount for these data. A plausible explanation for this can be made based on the different nature of the two approaches. In this synchronous wave of cells, there are a large number of cells just entering S-phase at the early times and leaving S-phase at later times. Since the pulse is 10 min, conceivably a significant number of cells could enter S-phase in that time and take up label, but the difference in DNA content compared to a G_1 cell would be insignificant. Flow cytometry would register it as a G_1 cell.

This would also happen to a wave of cells leaving S-phase into G_2 . From the initial slope of the curve in Figure 4, one can deduce that in a 10-min pulse, the percentage change in S-phase cells is about 4%. This is roughly the average difference between the labeling results and the computer analysis results.

A very difficult distribution for any model to fit accurately is that obtained when there are large numbers of cells in very late S. This situation is represented in Figures 3A-C. In these cases there is no apparent $G_2 + M$ peak, since it is overwhelmed by the late S-phase cells. A small shift in the location of the $G_2 + M$ peak can result in a large change in the S-phase percentage determined by the model, even though the fit may be nearly identical in both cases. Theoretically, this is also true for G_1 and early S cells, but the G_1 peak is so sharp in these data that it is present as a well-defined peak and can be fit accurately. Of course, this might not be the case with a low-resolution flow cytometer. These considerations apply irrespective of the model used to analyze the data. It is extremely important in these cases to know the G_2/G_1 ratio precisely, and to be able to fix the G_2 peak relative to the G_1 peak. The model allows for that option. The largest variation between the model results and labeling index indeed occur in these circumstances, though the difference is still less than ten percent. In light of these considerations, a researcher would be well-advised to do labeling index of cells if he desires

a precise analysis of S-phase when there are large numbers of late-S or early-S cells.

The use of a second-order polynomial plus a floating Gaussian (normal) curve is able to represent complex S-phase distributions, as shown here. Furthermore, the model makes physical sense. Initially the cells are synchronized in mitosis, presumably with a normal or near-normal distribution. As they move through the cell cycle, the normal distribution should still be apparent, but it will be distorted because of different rates of movement through the various phases and within each phase. The second-order polynomial allows enough freedom to fit distortions of the normal distribution. The experiments reported here to test the model are stringent tests using real data, and the results verify that the model can fit complex distributions with good results.

In summary, this paper has presented a new model to fit data from synchronous or asynchronous populations by fitting a smooth curve to S-phase. The model has been tested with a series of synchronized cultures by comparing the results to labeling index, with good results.

A copy of the Fortran code is available. The current version is written in an interactive mode for a CDC system, so some modifications might need to be made for another system.

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